Experiment 9

AIM: Use pytest to test python application.

1. Objective

- Understand unit testing concepts
- Write test cases using PyTest
- Execute and analyze test results
- Handle edge cases and exceptions
- Use parameterized testing

2. Prerequisites

- Python
- Pytest

3. Installation

Install pytest using pip:

```
pip install pytest
```

Verify it :

```
pytest –version
```

4. Create a Python module with basic mathematical operations and write PyTest test cases to validate:

- Correct outputs
- Edge cases
- Exception handling

Step 1: Create Project PythonDemo and add file pythondemo.py

```
def add(a, b):
 return a + b
```

```
def subtract(a, b):
 return a - b
def multiply(a, b):
 return a * b
def divide(a, b):
 if b == 0:
 raise ValueError("Cannot divide by zero")
 return a / b
def is_even(num):
 return num % 2 == 0
```

Step 2: Create a test file test_pythondemo.py

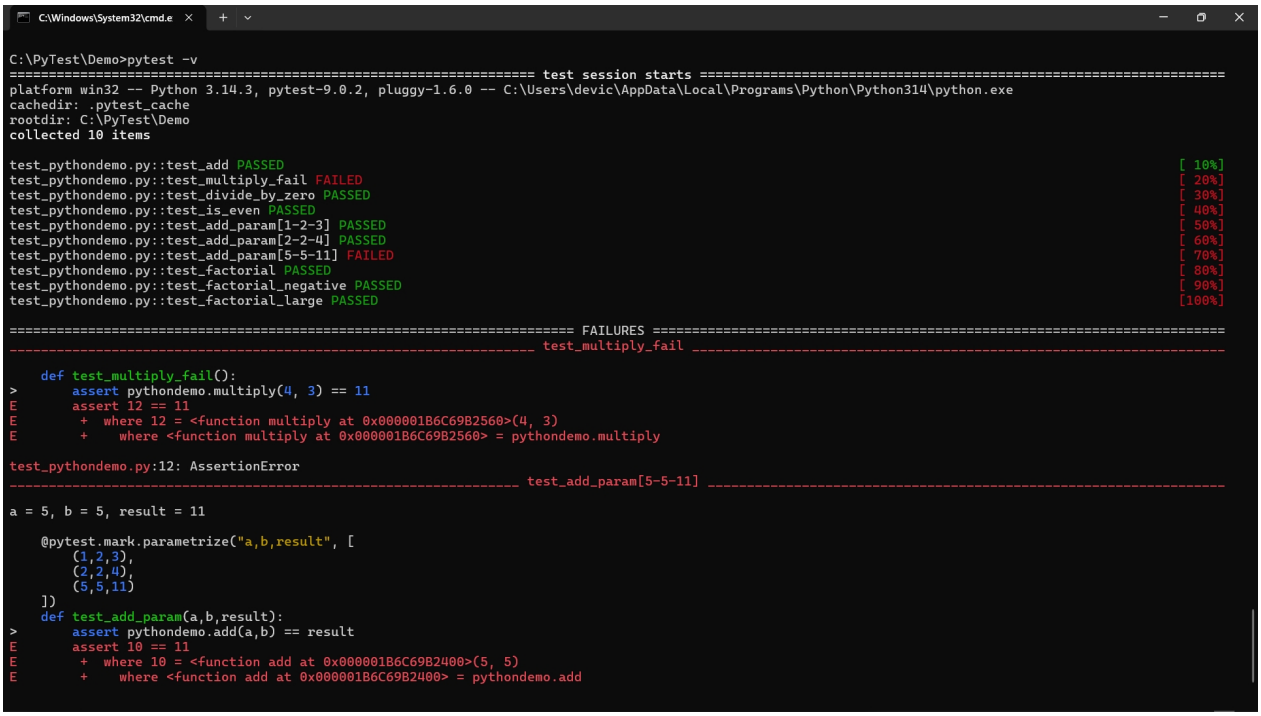
```
import pytest
import pythondemo
# Passing test
def test_add():
 assert pythondemo.add(2, 3) == 5
# Failing test
def test_multiply_fail():
 assert pythondemo.multiply(4, 3) == 11
# Exception test
def test_divide_by_zero():
 with pytest.raises(ValueError):
 pythondemo.divide(10, 0)
# Logical test
def test_is_even():
```

```
 assert pythondemo.is_even(4) is True

 assert pythondemo.is_even(5) is False

# Parametrized test

@pytest.mark.parametrize("a,b,result", [

 (1,2,3),

 (2,2,4),

 (5,5,11) # failing case

])

def test_add_param(a,b,result):

 assert pythondemo.add(a,b) == result
```

5. Execution

Open terminal in that folder

pytest –v



Task 1 and 2:

1. PythonDemp.py

```
def add(a, b):
    return a + b


def subtract(a, b):
    return a - b


def multiply(a, b):
    return a * b


def divide(a, b):
    if b == 0:
        raise ValueError("Cannot divide by zero")
    return a / b


def is_even(num):
    return num % 2 == 0



# Task 1 - factorial function
def factorial(n):
    if n < 0:
```

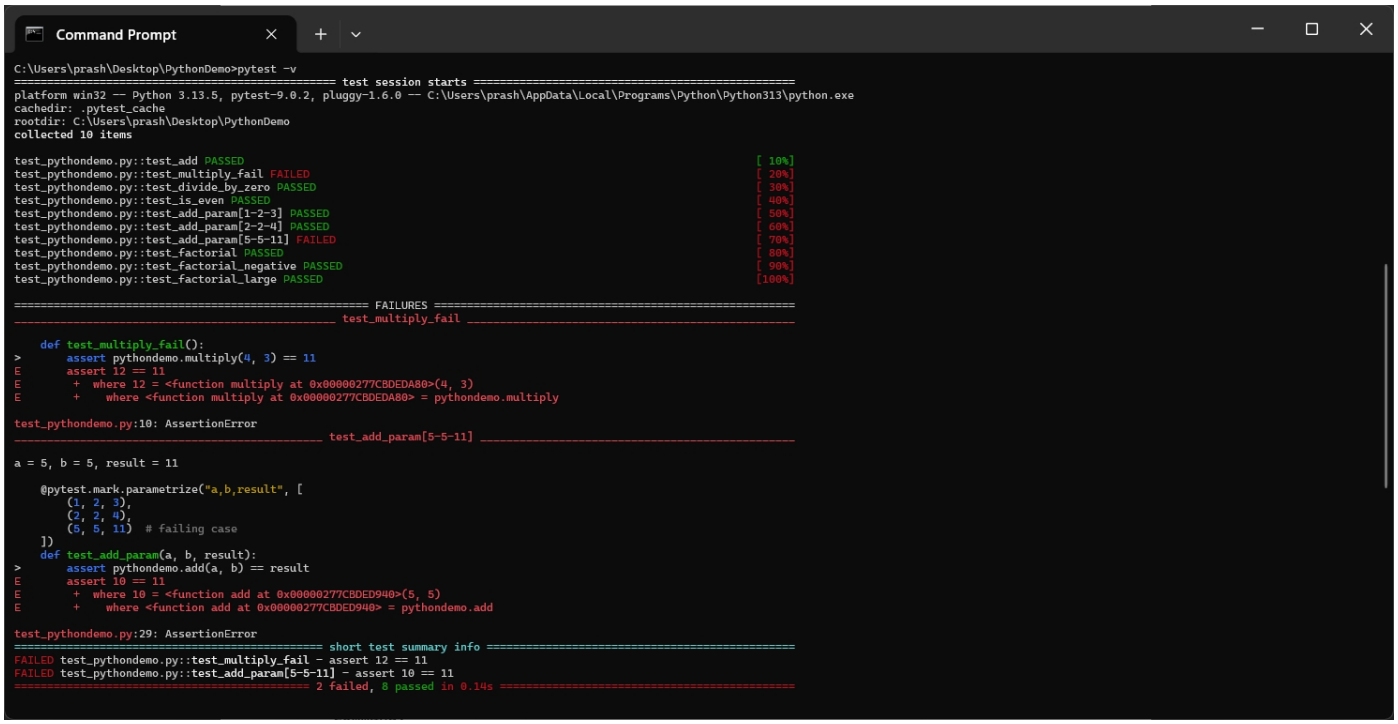
```
        raise ValueError("Negative numbers not allowed")


    result = 1
    for i in range(1, n + 1):
        result *= i

    return result
```

2. Test_PythonDemo.py

```
import pytest
import PythonDemo



# Passing test
def test_add():
    assert PythonDemo.add(2, 3) == 5



# Failing test
def test_multiply_fail():
    assert PythonDemo.multiply(4, 3) == 11



# Exception test
def test_divide_by_zero():
```

```
    with pytest.raises(ValueError):
        PythonDemo.divide(10, 0)


# Logical test
def test_is_even():
    assert PythonDemo.is_even(4) is True
    assert PythonDemo.is_even(5) is False



# Parameterized test
@pytest.mark.parametrize("a,b,result", [
    (1,2,3),
    (2,2,4),
    (5,5,11)   # failing case
])
def test_add_param(a,b,result):
    assert PythonDemo.add(a,b) == result



# Task 1 – factorial tests
def test_factorial():
    assert PythonDemo.factorial(5) == 120
    assert PythonDemo.factorial(0) == 1
```

```
# Task 2 – negative number test

def test_factorial_negative():

    with pytest.raises(ValueError):

        PythonDemo.factorial(-5)



# Task 2 – large input test

def test_factorial_large():

    assert PythonDemo.factorial(10) == 3628800
```




```
Command:

Microsoft Windows [Version 10.0.26200.8037]

(c) Microsoft Corporation. All rights reserved.



C:\Users\prash>pip install pytest
```

```
Collecting pytest
  Downloading pytest-9.0.2-py3-none-any.whl.metadata (7.6 kB)
Collecting colorama>=0.4 (from pytest)
  Using cached colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Collecting iniconfig>=1.0.1 (from pytest)
  Downloading iniconfig-2.3.0-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: packaging>=22 in
c:\users\prash\appdata\local\programs\python\python313\lib\site-packages (from
pytest) (26.0)
Collecting pluggy<2,>=1.5 (from pytest)
  Downloading pluggy-1.6.0-py3-none-any.whl.metadata (4.8 kB)
Collecting pygments>=2.7.2 (from pytest)
  Downloading pygments-2.20.0-py3-none-any.whl.metadata (2.5 kB)
Downloading pytest-9.0.2-py3-none-any.whl (374 kB)
Downloading pluggy-1.6.0-py3-none-any.whl (20 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading iniconfig-2.3.0-py3-none-any.whl (7.5 kB)
Downloading pygments-2.20.0-py3-none-any.whl (1.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.2/1.2 MB 9.2 MB/s eta 0:00:00
Installing collected packages: pygments, pluggy, iniconfig, colorama, pytest
Successfully installed colorama-0.4.6 iniconfig-2.3.0 pluggy-1.6.0 pygments-2.20.0
pytest-9.0.2

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
C:\Users\prash>pytest --version

pytest 9.0.2


C:\Users\prash>cd C:\Users\prash\Desktop


C:\Users\prash\Desktop>mkdir PythonDemo


C:\Users\prash\Desktop>cd PythonDemo


C:\Users\prash\Desktop\PythonDemo>notepad pythondemo.py


C:\Users\prash\Desktop\PythonDemo>notepad test_pythondemo.py


C:\Users\prash\Desktop\PythonDemo>pytest -v

============================================= test session starts
=============================================

platform win32 -- Python 3.13.5, pytest-9.0.2, pluggy-1.6.0 --
C:\Users\prash\AppData\Local\Programs\Python\Python313\python.exe

cachedir: .pytest_cache

rootdir: C:\Users\prash\Desktop\PythonDemo

collected 10 items


test_pythondemo.py::test_add PASSED
[ 10%]

test_pythondemo.py::test_multiply_fail FAILED
[ 20%]
```

```
test_pythondemo.py::test_divide_by_zero PASSED
[ 30%]

test_pythondemo.py::test_is_even PASSED
[ 40%]

test_pythondemo.py::test_add_param[1-2-3] PASSED
[ 50%]

test_pythondemo.py::test_add_param[2-2-4] PASSED
[ 60%]

test_pythondemo.py::test_add_param[5-5-11] FAILED
[ 70%]

test_pythondemo.py::test_factorial PASSED
[ 80%]

test_pythondemo.py::test_factorial_negative PASSED
[ 90%]

test_pythondemo.py::test_factorial_large PASSED
[100%]


================================================== FAILURES
==================================================

_____________________________________________ test_multiply_fail
_____________________________________________


    def test_multiply_fail():

>       assert pythondemo.multiply(4, 3) == 11

E       assert 12 == 11

E        +  where 12 = <function multiply at 0x00000277CBDEDA80>(4, 3)

E        +    where <function multiply at 0x00000277CBDEDA80> = pythondemo.multiply
```

```
test_pythondemo.py:10: AssertionError

_____________________________________________ test_add_param[5-5-11]
_____________________________________________


a = 5, b = 5, result = 11


    @pytest.mark.parametrize("a,b,result", [

        (1, 2, 3),

        (2, 2, 4),

        (5, 5, 11)  # failing case

    ])

    def test_add_param(a, b, result):

>       assert pythondemo.add(a, b) == result

E       assert 10 == 11

E        +  where 10 = <function add at 0x00000277CBDED940>(5, 5)

E        +    where <function add at 0x00000277CBDED940> = pythondemo.add


test_pythondemo.py:29: AssertionError

============================================= short test summary info
=============================================

FAILED test_pythondemo.py::test_multiply_fail - assert 12 == 11

FAILED test_pythondemo.py::test_add_param[5-5-11] - assert 10 == 11

=========================================== 2 failed, 8 passed in 0.14s
===========================================
```